



RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 9 (Summary and Tutorial)

Topic: Graphing Techniques and Transformations

Summary for Graphing Techniques

Recall using SIA as reference when sketching and labelling the curves you draw

S: Shape (including coordinates of stationary points if any)

I: Intercepts with the Axes (let $x = 0$ to find y , let $y = 0$ to find x)

A: Asymptotes (vertical, horizontal and oblique) – remember to label the equations of asymptotes.

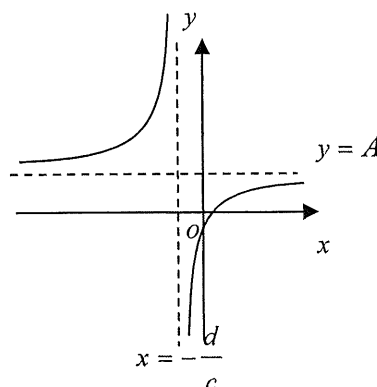
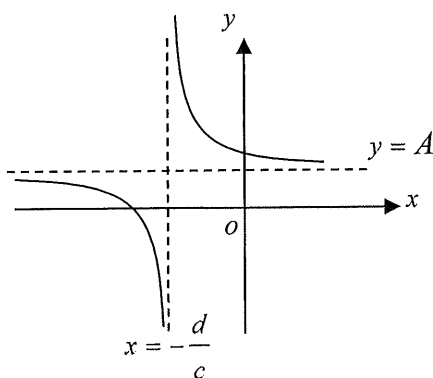
Consider to use the Graphing Calculator where applicable.

A. Rational functions

Type 1: $y = \frac{ax+b}{cx+d}$, $c \neq 0$, numerator and denominator have no common factor

$$y = \frac{ax+b}{cx+d} = A + \frac{B}{cx+d}$$

This is also known as rectangular hyperbola; the shape is either one of the following



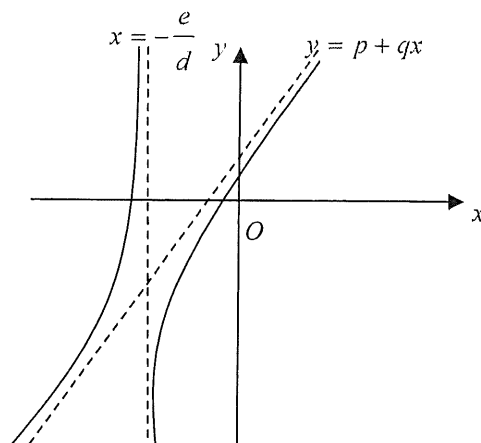
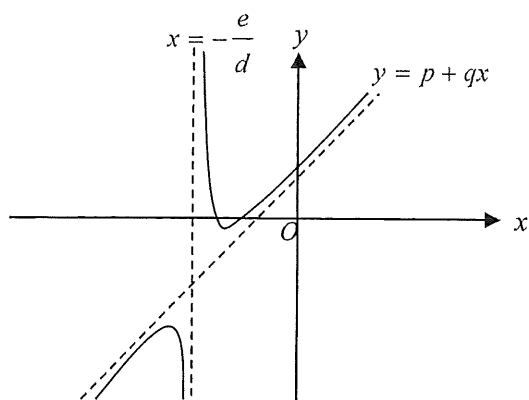
- Vertical asymptote : $x = -\frac{d}{c}$
- Horizontal asymptote : $y = A = \frac{a}{c}$

If $a = 0$, the horizontal asymptote is $y = 0$, i.e. x -axis

Useful to note: This graph has no turning points.

Type 2: $y = \frac{ax^2 + bx + c}{dx + e}$, $a \neq 0$, $d \neq 0$, numerator and denominator have no common factor

Note that $y = \frac{ax^2 + bx + c}{dx + e}$ can be re-written as $(p + qx) + \frac{r}{dx + e}$



- Vertical asymptote : $x = -\frac{e}{d}$
- Oblique asymptote : $y = px + q$

Example: $y = \frac{2x^2 + 5x + 3}{x + 2} \Rightarrow y = 2x + 1 + \frac{1}{x + 2}$

Asymptotes: $x = -2$ and $y = 2x + 1$

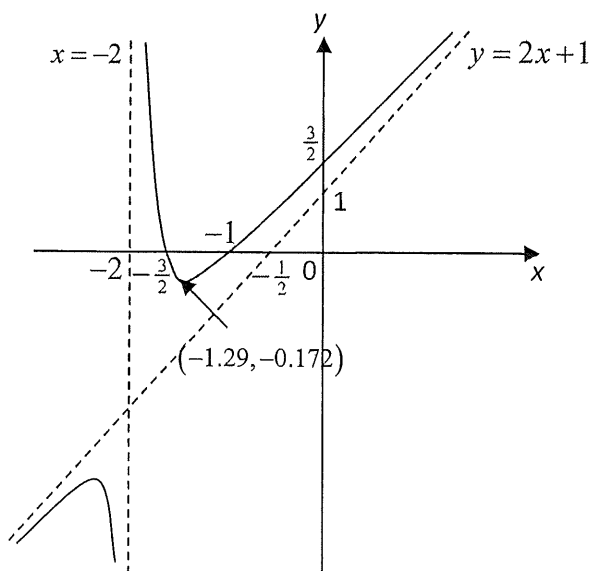
When $x = 0$, $y = \frac{3}{2}$

When $y = 0$,

$$2x^2 + 5x + 3 = 0$$

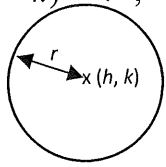
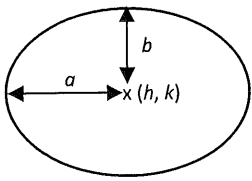
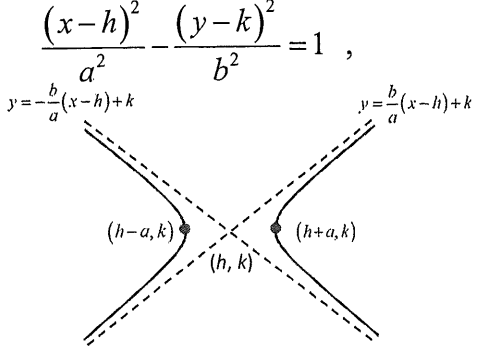
$$(2x + 3)(x + 1) = 0$$

$$x = -\frac{3}{2} \text{ or } -1$$



B. Conics

Note that you should use the **same scale for both axes**. If unsure about the shape of the curve, refer to the CONICS app in your GC. You can also use the app to sketch and/or check your answers.

Shape	Equation(s) in Standard Form	Point(s) to note when sketching
Circle	$(x-h)^2 + (y-k)^2 = r^2, \text{ where } r > 0.$ 	- Note that this is just a translation of the basic circle $x^2 + y^2 = r^2$. - Use a compass - Label the coordinates of the centre and radius clearly - Check if the origin is in the circle, on the circle, or outside the circle
Ellipse	$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1,$ where $a > 0, b > 0$. 	- Note that this is just a translation of the basic ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. - Graph is symmetrical about the lines $x = h$ and $y = k$ - Label the centre and lengths of semi-major and semi-minor axes clearly - Check if the origin is in the ellipse, on the ellipse, or outside the ellipse - If $a = b$, the equation becomes the equation of a circle with centre (h, k) and radius a units. - A circle is a special type of ellipse.
Hyperbola	$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1,$ 	- Note that this is just a translation of the basic hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. - Curve should be drawn such that it tends towards the oblique asymptotes - Label the centre and vertices of the hyperbola - Label the equations of the asymptotes - The oblique asymptotes intersect at the centre of the hyperbola, (h, k).

Shape	Equation(s) in Standard Form	Point(s) to note when sketching
		Finding equations of asymptotes Step 1: Rewrite the equation in the form $\frac{(y-k)^2}{b^2} = \frac{(x-h)^2}{a^2} - 1$. Step 2: Ignore the term '-1' since it is insignificant when $x \rightarrow \pm\infty$. So we have $\frac{(y-k)^2}{b^2} = \frac{(x-h)^2}{a^2}$. Step 3: Take square root on both sides of equation and add "$\pm$" on RHS of equation. So we have $\frac{y-k}{b} = \pm \frac{x-h}{a}$. Step 4: Proceed to make y the subject of the formula.

C. Using GC to sketch graphs of Parametric Equations

Step 1: Ensure that the mode is set to '**PARAMETRIC**'.

Step 2: Press $\boxed{y=}$ and enter the equations.

Step 3: Press $\boxed{\text{graph}}$ to visualize the graph. The default setting is $T_{\min} = 0, T_{\max} = 2\pi$.

Step 4: If the range of T is not between 0 and 2π inclusive, press $\boxed{\text{window}}$ and **adjust the values for $T_{\min}$ and $T_{\max}$ according to the domain of the parameter** as stated in the question.

D. Finding the Cartesian equation of Parametric Equations

The idea is to obtain a single equation involving the variables x and y only. This can often be done by eliminating the parameter via substitution method. However for parametric equations involving trigonometrical functions, you should consider using the trigonometric identities such as $\sin^2 x + \cos^2 x = 1$ and $\tan^2 x + 1 = \sec^2 x$ to formulate the Cartesian equation.

Important note on the use of GC

The GC gives the shape of the graph. In many cases, you **may need to resize the window (or Zoomfit) to capture the entire graph**, or to change window settings to capture critical features of the graph. We can use it to find intersections with axes and turning points, but **it does not draw asymptotes!** To determine if a graph has any asymptotes, you need to examine its equation.

Summary for Transformations**Standard Transformations**

The effects of standard transformations (translation, scaling and reflection) on the graph of $y = f(x)$ are summarised in the table below.

Note that a is taken to be a positive constant.

Replacement of x / y	Transformed Equation	Description of transformation (to be used when question requires description of transformations)
Replace x by $x - a$	$y = f(x - a)$	Translation of a units in the positive x -direction
Replace x by $x + a$	$y = f(x + a)$	Translation of a units in the negative x -direction
Replace y by $y - a$	$y - a = f(x)$ i.e. $y = f(x) + a$	Translation of a units in the positive y -direction
Replace y by $y + a$	$y + a = f(x)$ i.e. $y = f(x) - a$	Translation of a units in the negative y -direction
Replace x by $\frac{x}{a}$	$y = f\left(\frac{x}{a}\right)$	Scaling parallel to the x -axis by factor of a (in terms of coordinates, no change in corresponding y -coordinate)
Replace y by $\frac{y}{a}$	$\frac{y}{a} = f(x)$ i.e. $y = af(x)$	Scaling parallel to the y -axis by factor of a (in terms of coordinates, no change in corresponding x -coordinate)
Replace x by $-x$	$y = f(-x)$	Reflection in the y -axis
Replace y by $-y$	$-y = f(x)$ i.e. $y = -f(x)$	Reflection in the x -axis

Remarks:

For questions involving only standard transformations, there may be more than one sequence of transformations. For example, to transform $f(x)$ into $f(ax + b)$, you may

Either replace x by $x + b$, then replace x by ax

OR replace x by ax , then replace x by $x + \frac{b}{a}$.

However, for questions involving non-standard transformation(s) such as modulus, the sequence would be restricted. For example, to transform $f(x)$ into $f(-|x|)$, you can only

replace x by $-x$, then replace x by $|x|$.

Sketching the graph of $y = |f(x)|$ and $y = f(|x|)$ from the graph of $y = f(x)$

$y = f(x) = \begin{cases} f(x) & , \text{ if } f(x) \geq 0 \\ -f(x) & , \text{ if } f(x) < 0 \end{cases}$	$y = f(x) = \begin{cases} f(x), & x \geq 0, \\ f(-x), & x < 0. \end{cases}$
The graph of $y = f(x) $ lies above (or on) the x -axis.	The graph of $y = f(x)$ is symmetrical about the y -axis.
To obtain this graph from the graph of $y = f(x)$:	
1. Retain the part of the graph of $y = f(x)$ where $f(x) \geq 0$. 2. Reflect in the x -axis the part of the graph of $y = f(x)$ where $f(x) < 0$.	1. Retain the part of the graph of $y = f(x)$ where $x \geq 0$ 2. Reflect the part drawn in 1. in the y -axis so that the graph of $y = f(x)$ is symmetrical about the y -axis.

Sketching the graph of $y = f'(x)$ from the graph of $y = f(x)$

Note that we are sketching the **gradient** of $y = f(x)$, that is, we need to study the **gradient** of $y = f(x)$.

Regardless of whether $f(x) < 0$ (or > 0) in a particular interval, as long as the gradient is positive, $y = f'(x)$ will still be above the x -axis for that interval.

Similar argument follows if gradient is negative in an interval.

The following shows the details for your reference if necessary.

Graph of $y = f(x)$ (Note the description of the gradient)	Graph of $y = f'(x)$, that is graph of gradient of $y = f(x)$
Stationary point at $x = k$ (Gradient of stationary point is 0)	x -intercept at $x = k$
Strictly increasing over (a, b) (Gradient of graph is positive for (a, b))	Above x -axis over (a, b)
Strictly decreasing over (a, b) (Gradient of graph is negative for (a, b))	Below x -axis over (a, b)
Vertical asymptote $x = k$ (Gradient of vertical line is undefined)	Vertical asymptote $x = k$
Horizontal asymptote $y = k$ (Gradient of any horizontal line is 0)	Horizontal asymptote $y = 0$
Oblique asymptote $y = kx + c$ (Gradient of $y = kx + c$ is k)	Horizontal asymptote $y = k$

Sketching the graph of $y = \frac{1}{f(x)}$ from the graph of $y = f(x)$

We will still use SIA to guide us in sketching $y = \frac{1}{f(x)}$,

- identify any x -intercepts (vertical asymptotes) as it gives vertical asymptotes (x -intercepts).
- any turning point(s) for $f(x)$, as max point would become min point and vice versa.

Note that if $f(x) > 0$ (or < 0) over a certain interval, then $\frac{1}{f(x)} > 0$ (correspondingly < 0) for that interval.

The following shows the details for your reference if necessary.

SIA	Features of $y = f(x)$	$y = f(x)$	$y = \frac{1}{f(x)}$
Intercepts and Asymptotes	x -intercepts	$(a, 0)$	Vertical asymptote $x = a$
	Asymptotes	Vertical: $x = a$	x -intercept
		Horizontal: $y = b, b \neq 0$	Horizontal: $y = \frac{1}{b}$
		Horizontal: $y = 0$	$\frac{1}{f(x)} \rightarrow \pm\infty$
		Oblique: $y = ax + b$	Horizontal: $y = 0$
Shape	Turning points	Max pt (h, k)	Min pt $\left(h, \frac{1}{k}\right)$
		Min pt (h, k)	Max pt $\left(h, \frac{1}{k}\right)$
	y -values	(x, y)	$\left(x, \frac{1}{y}\right)$
		Above x -axis	Remains
		Below x -axis	Remains
		$f(x) = \pm 1$	$\frac{1}{f(x)} = \pm 1$
		$f(x)$ increases	$\frac{1}{f(x)}$ decreases
		$f(x)$ decreases	$\frac{1}{f(x)}$ increases
	Behaviour of graph	$f(x) \rightarrow +\infty$	$\frac{1}{f(x)} \rightarrow 0^+$
		$f(x) \rightarrow -\infty$	$\frac{1}{f(x)} \rightarrow 0^-$

Revision Tutorial Questions**Source of Question: AJC JC1 Promo 9758/2018/Q6(a)**

- 1 The curve C_1 has equation $y = f(x)$, where $f(x) = x^3 - 4x + 1$. A second curve C_2 undergoes, in succession, the following transformations to get the resulting curve C_1 :

A: A translation of 1 unit in the negative direction of the y -axis

B: A reflection in the y -axis

C: A scaling of factor $\frac{1}{4}$ parallel to the x -axis

Determine the equation of C_2 , showing your workings clearly.

[3]

Source of Question: ACJC JC2 Prelim 9758/2018/01/Q9b

- 2 The curve C has equation $y = \frac{3x^2 + 9x + 7}{x + 2}$.

(i) By expressing the equation of C in the form $y = 3x + a + \frac{b}{x+2}$, where a and b are non-zero constants, write down the equations of the asymptotes of C .

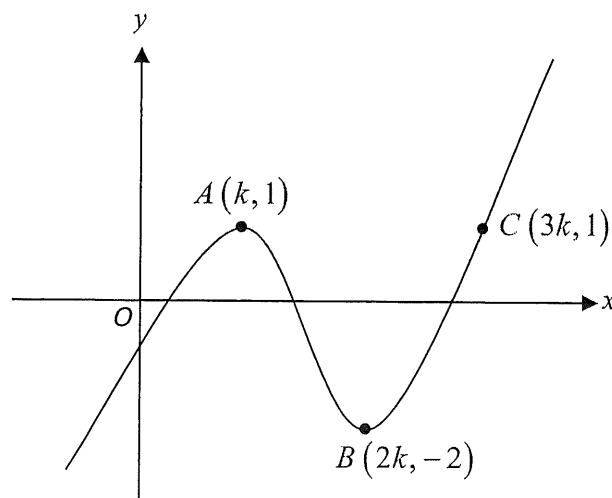
[2]

(ii) Describe a pair of transformations that will transform C to the graph of $y = 3x + \frac{1}{x}$.

[2]

Source of Question: MJC JC1 Promo 9740/2011/Q6a

- 3 The diagram below shows the graph of $y = f(x)$.



Sketch the graph of $y = f\left(\frac{1}{2}x + k\right) - 1$, indicating the coordinates of the points corresponding to A , B and C .

[3]

Source of Question: MJC JC1 Promo 9740/2012/Q5a

4 It is given that

$$g(x) = \begin{cases} 6 - x^2 & \text{for } 0 < x \leq 2, \\ 2x - 2 & \text{for } 2 < x \leq 4, \end{cases}$$

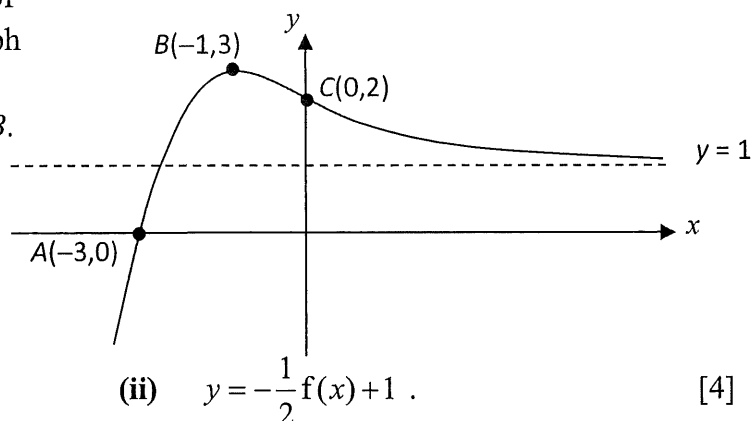
and that $g(x) = g(x+4)$ for all real values of x .

(i) Sketch the graph of $y = g(x)$ for $-6 \leq x \leq 10$. [3]

(ii) Evaluate $g(-3) + g(7)$ [2]

Source of Question: MI JC2 Promo 9740/2009/Q5

5 The diagram shows a sketch of the graph of $y = f(x)$. The graph cuts the axes at point A and C and has a maximum point at B . The equation $y = 1$ is an asymptote of the graph. On separate diagrams, sketch the graph of

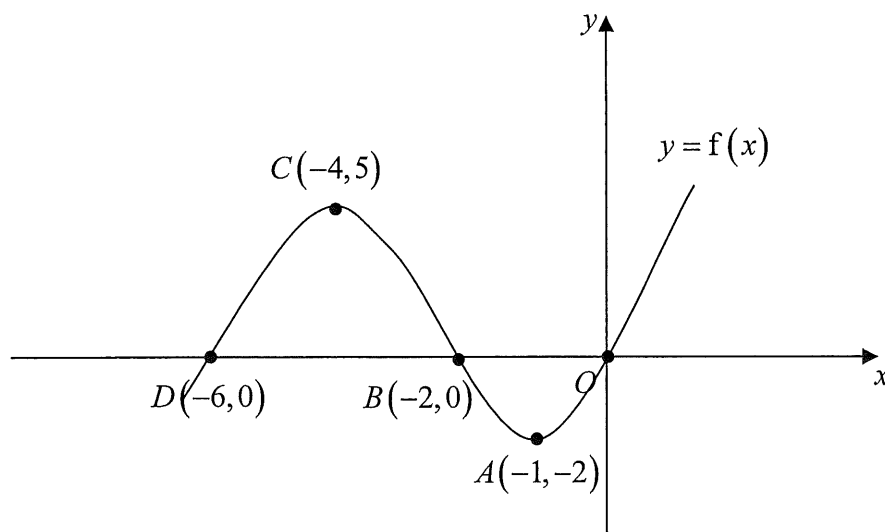


(i) $y = f(2x+1)$,

(ii) $y = -\frac{1}{2}f(x) + 1$. [4]

Source of Question: VJC JC2 Prelim 9758/2018/01/Q2

6



The diagram shows the graph of $y = f(x)$. The curve passes through the points $A(-1, -2)$, $B(-2, 0)$, $C(-4, 5)$, $D(-6, 0)$ and the origin O . Points A and C are the minimum point and maximum point respectively. On separate diagrams, sketch each of the following graphs indicating where possible, the coordinates of the stationary points, the points of intersections with the axes and the equations of any asymptotes.

(i) $y = f\left(\frac{x}{2} + 2\right)$ [2]

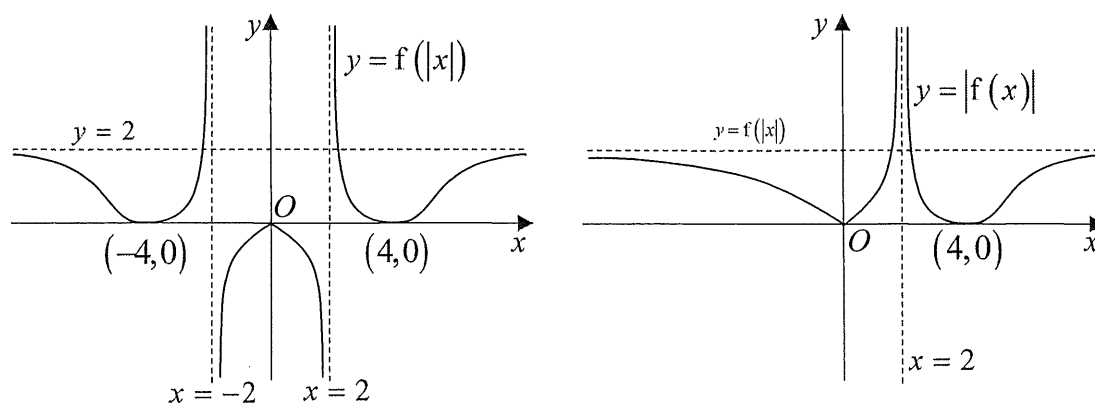
(ii) $y = \frac{1}{f(x)}$. [3]

Source of Question: SRJC JC2 Prelim 9758/2018/01/Q4

- 7 (i) The curve with equation $\frac{y^2}{9} - x^2 = 1$ undergoes a two-step transformations to become curve C with equation $\frac{y^2}{9} - \frac{(x-1)^2}{4} = 1$. [2]
- (ii) Draw a sketch of the curve C , labelling clearly the equation(s) of its asymptote(s), intersection with the axes and the coordinates of any turning points. [3]
- (iii) Show that the point $(-1, -3)$ lies on the line $y = mx + m - 3$ for all real values of m . [1]
- (iv) Hence using the diagram drawn in (ii), find the range of values of k such that the equation $\frac{(kx+k-3)^2}{9} - \frac{(x-1)^2}{4} = 1$ has 2 negative real roots. [2]

Source of Question: NJC JC2 Prelim 9758/2018/01/Q8b

- 8 The diagrams below show the graphs of $y = f(|x|)$ and $y = |f(x)|$, where the equations of the asymptotes and the coordinates of the turning points are given. The gradient of the graph with equation $y = f(x)$ at the origin is $-\frac{3}{4}$.



On separate diagrams, sketch the graphs of

- (i) $y = f(x)$, [2]
- (ii) $y = \frac{1}{f(x)}$, [3]
- (iii) $y = f'(x)$, [3]

including the coordinates of the points where the graphs cross the x - and y - axes and the equations of the asymptotes, if any.

Source of Question: AJC JC2 Prelim 9758/2018/02/Q3

- 9 The curve C has equation $y = ax + b + 2a + \frac{2(b+2a)}{x-2}$, where a and b are real constants such that $a > 0$, $b \neq -2a$ and $x \neq 2$.
- (i) By using differentiation, find the range of values of b in terms of a such that C has no stationary points. [4]
- (ii) Given that $b = -3a$, sketch C , stating the equations of any asymptotes and the coordinates of the points where C crosses the axes. [3]
- (iii) On the same diagram as C , sketch the graph of $\left(\frac{y}{a}\right)^2 - (x-1)^2 = 4$, showing clearly the equations of the asymptotes and turning points. Hence solve the inequality $x-1 - \frac{2}{x-2} \geq \sqrt{(x-1)^2 + 4}$. [4]

Source of Question: NJC JC2 Prelim 9758/2018/01/Q5

- 10 Given that a is a positive constant. A curve C_1 has parametric equations

$$x = \frac{a}{t}, \quad y = 1 + t.$$

Sketch C_1 , labelling the coordinates of the point(s) where the curve crosses the x - and y -axes, and the equations of the asymptote(s) in terms of a , if any. [2]

Another curve C_2 has equation $y = \sqrt{1 + \frac{x^2}{a^2}}$.

- (i) Show algebraically that the y -coordinates of the point(s) of intersection of C_1 and C_2 satisfies the equation $(y-1)^2(y^2-1)-1=0$. [2]
- (ii) Sketch C_2 on the same diagram as C_1 , labelling the coordinates of the point(s) where the curve crosses the x - and y -axes, and the equations of the asymptotes in terms of a , if any.
Find the coordinates of point(s) of intersections of C_1 and C_2 and label the coordinates in this diagram, leaving the answers correct to 3 significant figures, in terms of a . [4]

